

Category	Component	Model / Hardware	Qty	Main Spec	Connection / Port	Speed / Bandwidth	Current Status	Note / Manual
System	Machine name	BlackyMan / DESKTOP-2F4MA89	1	ASUS Z10PE-D8 WS workstation	Workstation / ATX full tower	Legacy PCIe Gen 3 platform	Active troubleshooting machine	Master spec for BlackyMan hardware baseline; Manual Box: https://sittichaiwong01.github.io/Manual/
Mainboard	Motherboard	ASUS Z10PE-D8 WS	1	Intel C612 workstation board; BIOS 4301	LGA2011-v3 dual CPU board	PCIe Gen 3 platform	Active; BIOS 4301 installed	Mainboard manual: https://sittichaiwong01.github.io/Manual/mobo-e9680_z10pe-d8_ws_ug_v2.pdf
BIOS	BIOS firmware	ASUS Z10PE-D8 WS BIOS 4301	1	Current BIOS baseline after upgrade	UEFI/BIOS setup	N/A	Current baseline	After Clear CMOS / Load Defaults, must verify Samsung 990 EVO M.2 2TB visibility before calling setup complete.
CPU	Processor	Intel Xeon E5-2650 v3	2	Dual Xeon Haswell-EP; 10 cores per CPU / 20 threads per CPU; base 2.30GHz	CPU Socket 1 + CPU Socket 2	QPI / DDR4 memory controller per CPU	Installed	Dual CPU memory population must be balanced for RAM training.
RAM	Memory target	Samsung 32GB DDR4 ECC Registered RDIMM	8	Target 256GB total; 8 x 32GB ECC RDIMM; DDR4-2133 class	DIMM slots A1-D1 / E1-H1	CPU1 side + CPU2 side memory channels	One DIMM/slot issue under investigation; current blocker Q-Code B7	Training order: A1+E1; then A1+B1+E1+F1; then A1+B1+C1+E1+F1+G1; then full A1-D1 + E1-H1.
RAM Training	Initial test pair	A1 + E1	1	Minimum dual CPU balanced RAM training pair	CPU1 A1 + CPU2 E1	N/A	Required first checkpoint	If A1+E1 fails, stop and isolate RAM stick / slot / CPU memory path / contact / socket pressure.
GPU	Current GPU	NVIDIA GeForce RTX 5060 Ti 16GB	1	16GB GDDR7; PCIe Gen 5 x8 class GPU	PCIe x16 physical slot; electrical x8 GPU design	Target on BlackyMan: PCIe 3.0 x8 stable	Compatibility issue under test: B2/WHEA17/freeze background	Do not use as first display path during RAM training. Goal is internal PCIe 3.0 x8 if stable.
GPU	Fallback GPU	ASUS GeForce GTX TITAN X 12GB	1	12GB VRAM; older stable NVIDIA GPU	PCIe x16 slot	PCIe Gen 3 generation GPU	Fallback/stable display/GPU option	Use if ASPEED output is unavailable and RTX 5060 Ti should be removed from variables.
Onboard Graphics	Onboard VGA	ASPEED AST2400 32MB	1	Onboard VGA controller for setup/server-style display	VGA via bracket / onboard VGA path	Up to 1920x1200 @ 60Hz per manual	Useful for BIOS/RAM training if enabled	Jumper VGA_SW1 controls onboard VGA. Enable = pins 1-2 according to manual.
Jumper	Onboard VGA jumper	VGA_SW1	1	Enable/disable onboard VGA controller	Motherboard jumper	N/A	Must verify before using ASPEED	Set pins 1-2 to activate VGA feature. Useful to reduce GPU compatibility variables.
Cooling	CPU cooler	Corsair H80i GT liquid CPU cooler	2	AIO liquid coolers for dual CPUs	CPU cooler mounts / fan/pump power	N/A	Installed	Do not change cooler pressure randomly during RAM troubleshooting unless memory path/socket pressure becomes the next hypothesis.
Cooling	Fan controller	NZXT SEN-MIX2 Sentry Mix 2 Fan Controller	1	Manual fan control; Sentry Mix 2	5.25-inch bay / fan channels	6 channels; 30W per channel class	Installed	Important hardware-control device; do not disable/remove blindly.
Cooling	Case fan	Cooler Master SickleFlow 120	5	120mm case fans	Connected to fan controller / motherboard as installed	N/A	Installed	Fan routing should stay stable during RAM training.
Power	Power supply	EVGA SuperNOVA 1600 G2 / 120-G2-1600-X1	1	1600W 80 PLUS Gold PSU	ATX power + PCIe power connectors	1600W capacity	Installed	PSU capacity is enough for RTX 5060 Ti / TITAN X context; still verify PCIe power connectors before GPU testing.
Storage	NVMe SSD	Samsung 990 EVO NVMe M.2 2TB	1	Main high-speed 2TB NVMe drive	M.2 / PCIe storage path	NVMe / PCIe	Must be visible after BIOS reset	Mandatory checklist after Clear CMOS: BIOS must see 2TB Samsung M.2 SSD as standard.
Storage	SATA SSD	Samsung 850 EVO 500GB	1	500GB SATA SSD	SATA	SATA 6Gbps class	Installed	Storage device; not required during first RAM training POST tests.
Storage	SATA SSD	Samsung 850 PRO 512GB	1	512GB SATA SSD	SATA	SATA 6Gbps class	Installed	Storage device; not required during first RAM training POST tests.
Network	Onboard LAN	Intel onboard Gigabit LAN	1	Onboard Intel 1GbE LAN	RJ-45 Ethernet	1GbE	Installed	Direct LAN 1GbE is current baseline; 10GbE is future option, not default.

Network	USB Wi-Fi	TP-Link Archer TX20U Plus	1 USB Wi-Fi adapter; 5GHz stable baseline	USB-A	Wi-Fi 6; adapter stable with USB Switch Mode = USB 2.0	Stable after setting USB Switch Mode to USB 2.0	Disconnect during RAM training to reduce variables.
USB	Onboard USB	ASUS Z10PE-D8 WS onboard USB ports	1 USB 3.x / USB 2.0 onboard ports	Rear/front USB headers and ports	USB 3.x class	Installed	During RAM training, keep only keyboard/mouse if needed.
USB Expansion	PCIe USB card	USB 3.2 Gen2x2 20Gbps PCIe card	1 High-speed USB expansion card	PCIe expansion slot	20Gbps class	Installed / available	Not required for RAM training. Remove/ignore if reducing PCIe variables.
Optical	DVD drive	ASUS 24X DVD Burner	1 DVD-RW drive	SATA	N/A	Installed	Not required for RAM training.
Case	Case	Phanteks Enthoo Pro ATX Full Tower Case	1 Full tower case	ATX / workstation case	N/A	Installed	Large case suitable for workstation hardware and cooling.
OS	Operating system	Windows 10 Pro	1 Windows 10 Pro baseline	Boot drive on internal storage	N/A	Installed	Do not troubleshoot Windows until RAM training/POST baseline is stable.
Manual Source	Manual Box / GitBox	Manual repository	1 User-provided hardware/manual reference box	Web URL	N/A	Primary reference source	Use this first for manuals and hardware spec: https://sittichaiwong01.github.io/Manual/
Manual Source	BlackyMan spec Excel	BlackymanSpec.xlsx	1 User hardware spec spreadsheet reference	Web URL	N/A	Reference file	Spec source: https://sittichaiwong01.github.io/Manual/BlackymanSpec.xlsx